

Restaurant Discovery System

Software Requirement Specification

Module: Restaurant Discovery
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Version 1.5
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Restaurant Discovery System module of a web-based food review and restaurant advertising platform. This document is intended for use by the development teams and the course evaluator.

1.2 Scope

1.2.1 Scope

The Restaurant Discovery System enables users to find, filter, and interact with restaurant listings. The system covers four main capability areas:

- Search & Filtering — keyword search and multi-dimensional filtering of restaurant listings
- Location Services — map-based discovery and proximity-based restaurant recommendations
- Basic User Interaction — saving favorites and sharing restaurant listings

This module does not cover restaurant owner advertising management, review submission, or user account authentication, which are handled by separate modules.

1.2.2 Out of Scope

Due to this being version 1.0, the following features are explicitly excluded from the current development cycle of the Restaurant Discovery System module, but may be implemented in future versions:

- **Social Feed & User Profiles:** Following other users, viewing public user profiles, or sharing listings to an internal social activity feed are excluded from this module. Sharing is strictly limited to copying direct URLs to the device clipboard.
- **Table Bookings & Reservations:** Real-time seat availability checking, calendar integration, and reservation placements are outside the scope of the discovery system.

1.3 Definitions and Abbreviations

Term / Abbreviation	Definition
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UI	User Interface
API	Application Programming Interface
GPS	Global Positioning System / browser geolocation
Guest User	An unauthenticated visitor browsing the website
Registered User	A user with an account who is logged in
Listing	A restaurant entry in the database

1.4 Document Overview

Section 2 describes the overall system context and assumptions. Section 3 details all functional requirements organized by feature group. Section 4 lists non-functional requirements. Section 5 presents the use case summary table.

2. Overall Description

2.1 System Context

The website is intended to be a full-stack web application. The Restaurant Discovery System is a front-end and back-end module that interacts with a database of restaurant listings. It also integrates with the browser Geolocation API and a third-party mapping service (e.g., Google Maps or Leaflet/OpenStreetMap) for location-based features.

2.2 User Classes

Guest User: Can search, filter, view listings, view map, and share links. Cannot save favorites.

Registered User: All guest capabilities plus: save/unsave restaurants to a personal favorites list.

Restaurant Owner: May appear in search/filter results but does not have special discovery-module interactions.

2.3 Assumptions and Dependencies

- The database contains restaurant records with name, cuisine type, location (address and/or coordinates), price range, and aggregate rating.
- A mapping library (Google Maps API or Leaflet + OpenStreetMap) is available and licensed for use.
- The browser environment supports the HTML5 Geolocation API.
- Users who deny location permissions must still be able to use the search and filter features without location-based results.

3. Functional Requirements

Requirements are identified using the format FR-DS-XX-### where DS indicates the Discovery System module and XX indicates the sub-module. Priority levels are High, Medium, or Low.

3.1 Search & Filtering

3.1.1 Keyword Search

Users shall be able to type a search query to find restaurants by name or relevant keyword (e.g., cuisine style, dish name).

Req. ID	Feature	Description	Priority
FR-DS-SF-001	Keyword Search Input	The system shall provide a text input field on the main discovery page where users can enter a search query.	High
FR-DS-SF-002	Search by Name	The system shall return restaurant listings whose name contains the search string (case-insensitive, partial match).	High
FR-DS-SF-003	Search by Keyword	The system shall return listings whose associated tags, cuisine type, or description contain the search keyword.	High
FR-DS-SF-004	Empty Search State	When no query is entered, the system shall display all available listings (subject to active filters and sort order).	Medium
FR-DS-SF-005	No Results State	When a search yields no results, the system shall display a clear 'No restaurants found' message with a suggestion to broaden the search.	Medium

3.1.2 Filter by Cuisine Type

Req. ID	Feature	Description	Priority
FR-DS-SF-006	Cuisine Filter Options	The system shall provide a selectable list of cuisine types (e.g., Italian, Japanese, Mexican, Vegan) derived from the database.	High
FR-DS-SF-007	Multi-Select Cuisine Filter	Users shall be able to select one or more cuisine types simultaneously to narrow results.	Medium
FR-DS-SF-008	Cuisine Search Auto-complete	As the user types in the cuisine filter search box, the system should suggest matching cuisine tags available in the database.	Low

3.1.3 Filter by Location

Req. ID	Feature	Description	Priority
FR-DS-SF-09	Location Text Filter	Users shall be able to type a neighbourhood, city, or postal code to filter listings by location text.	Medium

3.1.4 Filter by Price Range

Req. ID	Feature	Description	Priority
FR-DS-SF-010	Price Range Selector	The system shall provide a price range filter using a standard \$ / \$\$ / \$\$\$ / \$\$\$\$ notation.	High
FR-DS-SF-011	Multi-Select Price	Users shall be able to select multiple price tiers simultaneously.	Low

3.1.5 Filter by Rating

Req. ID	Feature	Description	Priority
FR-DS-SF-012	Minimum Rating Filter	The system shall provide a minimum star rating filter (e.g., 3+, 4+, 5 stars only).	High
FR-DS-SF-013	Rating Display	Each listing in search results shall display its current aggregate star rating.	High

3.1.6 Sort Results

Req. ID	Feature	Description	Priority
FR-DS-SF-014	Sort by Rating	Users shall be able to sort results in descending order of aggregate rating.	High
FR-DS-SF-015	Sort by Distance	When geolocation is active, users shall be able to sort results by distance (nearest first).	Medium
FR-DS-SF-016	Sort by Newest	Users shall be able to sort results by listing creation date, newest first.	Low
FR-DS-SF-017	Default Sort Order	The default sort order shall be by rating (descending) unless the user has modified it.	Low
FR-DS-SF-018	Sort by Price	Users shall be able to sort the filtered restaurant results by price range (from cheapest to most expensive).	Low

3.1.7 Master Filter Reset

Req. ID	Feature	Description	Priority
FR-DS-SF-019	Global Clear All Filters	The system shall provide a visible "Clear All" or "Reset Filters" button that clears all active cuisine, price, location, and location filters simultaneously, instantly reverting the search results to the default display state.	High

3.2 Location Services

3.2.1 Map View

Req. ID	Feature	Description	Priority
FR-DS-LS-020	Map Display	The system shall render an interactive map with pin markers representing restaurant locations matching the current search/filter state.	High
FR-DS-LS-021	Map-Listing Synchronization	Selecting a pin on the map shall highlight the corresponding listing card in the results panel, and vice versa.	Low
FR-DS-LS-022	Map Zoom & Pan	Users shall be able to zoom and pan the map; the visible listing results shall update dynamically to reflect the current map viewport.	Low
FR-DS-LS-023	Marker Info Popup	Tapping or clicking a map pin shall display a small popup with the restaurant name, rating, and a link to the full listing.	Low

3.2.2 Restaurants Near Me

Req. ID	Feature	Description	Priority
FR-DS-LS-024	Geolocation Request	The system shall include a 'Near Me' button that triggers a browser geolocation permission request.	High
FR-DS-LS-025	Proximity Filter	Once location is granted, the system shall filter and sort results by distance from the user's current coordinates.	High
FR-DS-LS-026	Permission Denied Handling	If the user denies location access, the system shall display an informative message and revert to the default view without disrupting other features.	High
FR-DS-LS-027	Distance Display	When 'Near Me' is active, each listing card shall display the approximate distance from the user.	Medium

3.3 Basic User Interaction

3.3.1 Save to Favorites

Req. ID	Feature	Description	Priority
FR-DS-UI-028	Bookmark Control	Each listing card and detail page shall include a bookmark/heart icon control to save or unsave the restaurant.	Medium
FR-DS-UI-029	Authentication Gate	If a guest user attempts to save a restaurant, the system shall prompt them to log in or register before proceeding.	High
FR-DS-UI-030	Favorites Persistence	Saved restaurants shall persist across sessions for registered users.	High
FR-DS-UI-031	Visual Saved State	The bookmark icon shall visually indicate whether a restaurant is already saved (filled vs. outline state).	Medium
FR-DS-UI-032	Favorites List Access	Registered users shall be able to navigate to a dedicated Favorites page listing all their saved restaurants.	Medium

3.3.2 Share a Restaurant Listing

Req. ID	Feature	Description	Priority
FR-DS-UI-033	Share Button	Each listing card and detail page shall include a 'Share' button or icon.	High
FR-DS-UI-034	Copy Link Action	Activating the Share button shall copy the direct URL of the restaurant listing to the user's clipboard.	Low
FR-DS-UI-035	Copy Confirmation	The system shall display a brief success notification (e.g., 'Link copied!') after the URL is copied.	Medium
FR-DS-UI-036	Guest Sharing	Share functionality shall be available to both guest and registered users without requiring authentication.	High

4. Non-Functional Requirements

NFR ID	Category	Requirement	Priority
NFR-DS-001	Performance	Search results shall load and render within 2 seconds under normal network conditions (broadband).	High
NFR-DS-002	Performance	Map tiles and pins shall render within 3 seconds on initial load.	Low
NFR-DS-003	Usability	The search bar, filters, and sort controls shall be accessible from a single page without requiring navigation away from the results list.	High
NFR-DS-004	Usability	The interface shall be responsive and usable on desktop, tablet, and mobile screen sizes.	Low
NFR-DS-005	Security	User location data obtained via geolocation shall not be stored on the server; it shall only be used client-side for the duration of the session.	Low
NFR-DS-006	Accessibility	Interactive controls (search, filters, buttons) shall have appropriate ARIA labels to support screen reader access.	Medium
NFR-DS-007	Scalability	The search and filter backend shall handle up to 10,000 restaurant listings without degraded response times.	Medium
NFR-DS-008	Compatibility	The website shall function correctly on latest versions of Chrome, Firefox, Edge etc.	Medium
NFR-DS-009	Usability	The system shall provide clear and user-friendly error messages when invalid input or system errors occur.	High

5. Use Case Summary

5.1 Search for Restaurants

- **Use Case ID:** UC-DS-01
- **Primary Actors:** Guest User, Registered User.
- **Description:**
 - Allows users to search for restaurants using keywords such as restaurant name, cuisine type, menu item, or location.
- **Goal:** Find restaurants matching user criteria.
- **Trigger:** User enters a search term and initiates a search.
- **Preconditions:**
 - System is available.
 - Restaurant database contains listings.
- **Main Success Scenario:**
 1. User accesses the search page.
 2. User enters a keyword.
 3. User clicks Search.
 4. System validates the input.
 5. System searches the database.
 6. System retrieves matching restaurants.
 7. System displays ranked search results.
 8. User selects a restaurant to view details.
- **Alternative Flows:**
 - Empty search field → System prompts user to enter a keyword.
 - No results found → System displays a "No Results Found" message.
- **Postconditions:** Matching restaurants are displayed.
- **Priority:** **High**

5.2 Apply Cuisine Filter

- **Use Case ID:** UC-DS-02
- **Primary Actors:** Guest User, Registered User.
- **Description:**
 - Allows users to narrow restaurant results by selecting one or more cuisine types.
- **Goal:** Help users find restaurants matching preferred cuisines.
- **Trigger:** User selects one or more cuisine filters.
- **Preconditions:**
 - Restaurant results are available.
 - Cuisine categories exist in the system.
- **Main Success Scenario:**
 1. User views restaurant search results.
 2. User selects one or more cuisine types.
 3. System applies the selected filters.
 4. System updates the restaurant list.
 5. User views filtered results.
- **Alternative Flows:** No restaurants match selected cuisines → System displays a notification.
- **Postconditions:** Filtered restaurant results are displayed.
- **Priority:** **Medium**.

5.3 Sort Results

- **Use Case ID:** UC-DS-03
- **Primary Actors:** Guest User, Registered User.

- **Description:**
 - Allows users to reorder restaurant results using sorting options such as rating, distance, popularity, or newest listings.
- **Goal:** Present restaurant results according to user preferences.
- **Trigger:** User selects a sorting option.
- **Preconditions:**
 - Restaurant results are displayed.
- **Main Success Scenario:**
 1. User opens the sort menu.
 2. User selects a sorting criterion.
 3. System reorders the restaurant results.
 4. Updated results are displayed.
- **Alternative Flows:** Invalid sorting option selected → System applies default sorting.
- **Postconditions:** Results are displayed in the selected order.
- **Priority:** **Medium.**

5.4 View Map

- **Use Case ID:** UC-DS-04
- **Primary Actors:** Guest User, Registered User.
- **Description:**
 - Allows users to view restaurant locations on an interactive map.
- **Goal:** Provide a geographic view of restaurant locations.
- **Trigger:** User selects the Map View option.
- **Preconditions:**
 - Restaurants have location data.
- **Main Success Scenario:**
 1. User clicks Map View.
 2. System loads the map interface.
 3. System displays restaurant markers.
 4. User interacts with map markers to view restaurant information.
- **Alternative Flows:** Map service unavailable → System displays an error message.
- **Postconditions:** Restaurant locations are shown on the map.
- **Priority:** **High.**

5.5 Use "Near Me"

- **Use Case ID:** UC-DS-05
- **Primary Actors:** Guest User, Registered User.
- **Description:**
 - Allows users to find restaurants located near their current location.
- **Goal:** Provide nearby restaurant recommendations.
- **Trigger:** User selects the "Near Me" feature.
- **Preconditions:**
 - User device supports location services.
 - User grants location permission.
- **Main Success Scenario:**
 1. User clicks "Near Me."
 2. System requests location access.
 3. User grants permission.
 4. System retrieves the user's location.
 5. System calculates nearby restaurants.

- 6. Results are sorted by distance.
- 7. Nearby restaurants are displayed.
- **Alternative Flows:** User denies location permission → System requests manual location input.
- **Postconditions:** Nearby restaurants are displayed.
- **Priority:** **High.**

5.6 Save Restaurant

- **Use Case ID:** UC-DS-06
- **Primary Actors:** Registered User.
- **Description:**
 - Allows registered users to save restaurants to a personal favorites list.
- **Goal:** Enable quick access to preferred restaurants.
- **Trigger:** User clicks the Save or Bookmark icon.
- **Preconditions:**
 - User is logged in.
 - Restaurant listing exists.
- **Main Success Scenario:**
 1. User views a restaurant listing.
 2. User clicks Save.
 3. System adds the restaurant to Favorites.
 4. System confirms the action.
- **Alternative Flows:** Restaurant already saved → System informs the user.
- **Postconditions:** Restaurant is stored in the user's Favorites list.
- **Priority:** **High.**

5.7 Share Restaurant Link

- **Use Case ID:** UC-DS-07
- **Primary Actors:** Guest User, Registered User.
- **Description:**
 - Allows users to share a restaurant listing using a generated link.
- **Goal:** Facilitate sharing restaurant information with others.
- **Trigger:** User selects the Share option.
- **Preconditions:**
 - Restaurant listing exists.
- **Main Success Scenario:**
 1. User clicks Share.
 2. System generates a shareable URL.
 3. System copies the URL to the clipboard or displays sharing options.
 4. System confirms the action.
- **Alternative Flows:** Clipboard access unavailable → System displays the URL manually.
- **Postconditions:** Shareable restaurant link is generated.
- **Priority:** **Low.**

5.8 View Favorites List

- **Use Case ID:** UC-DS-8
- **Primary Actors:** Registered User.
- **Description:**
 - Allows registered users to access all restaurants previously saved to their Favorites list.
- **Goal:** Provide easy access to bookmarked restaurants.

- **Trigger:** User navigates to the Favorites page.
- **Preconditions:**
 - User is logged in.
 - User has saved restaurants.
- **Main Success Scenario:**
 1. User opens the Favorites page.
 2. System retrieves saved restaurants.
 3. System displays the Favorites list.
 4. User selects a restaurant for more details.
- **Alternative Flows:** No saved restaurants exist → System displays an empty state message.
- **Postconditions:** Favorites list is displayed.
- **Priority:** **Medium.**

5.9 Compare Restaurants

- **Use Case ID:** UC-DS-9
- **Primary Actors:** Guest User, Registered User.
- **Description:**
 - Allows users to compare multiple restaurants side-by-side using key attributes.
- **Goal:** Help users make informed restaurant choices.
- **Trigger:** User selects two or more restaurants and chooses Compare.
- **Preconditions:**
 - At least two restaurants are selected.
- **Main Success Scenario:**
 1. User selects multiple restaurants.
 2. User clicks Compare.
 3. System retrieves restaurant information.
 4. System displays a side-by-side comparison including ratings, cuisine type, price range, location, and operating hours.
 5. User reviews the comparison.
- **Alternative Flows:** Fewer than two restaurants selected → System requests additional selections.
- **Postconditions:** Comparison view is displayed.
- **Priority:** **Low.**

6. Revision History

Version	Date	Author	Description
1.0	1 June 2026	Edrien Borigas	SRS Document Section Titles Template
1.0	1 June 2026	Ramanjeet Kaur	Introduction
1.1	2 June 2026	Vitumbiko Ngwira	Overall Description
1.1	2 June 2026	Edrien Borigas	SRS Document FR and NFR Tables
1.2	4 June 2026	Gursharn S. Sangha	Added: 1.2.1 out of scope 3.1.7 Master Filter Reset FR-DS-009 FR-DS-020
1.2	4 June 2026	Rajvir Kaur, Vitumbiko Ngwira, Ramanjeet Kaur	Use Case Summary Draft
1.3	4 June 2026	Rajvir Kaur, Vitumbiko Ngwira, Ramanjeet Kaur	Formatted Use Case Summary
1.4	5 June 2026	Kaur Rajvir	Added Restaurant comparison feature requirements and use case
1.4	5 June 2026	Kaur Rajvir	Added NFR-DS-009 AND 10
1.4	5 June 2026	Edrien Borigas, Gursharn S. Sangha, Rajvir Kaur, Vitumbiko Ngwira, Ramanjeet Kaur	Priority Check
1.5	7 June 2026	Gursharn S. Sangha	Updated Use Case Summary Changed priorities. Deleted Some features. Formated
1.5	7 June 2026	Edrien Borigas	Added Priority for NFR table